MEDIBYTES ARD | PROJECT REVIEW

UI and Workflow Verification Report

Baseline, test evidence, and scope after the instruction to preserve the existing UI

FINAL SCOPE	VALIDATION
No application UI or workflow source changes	**Frontend build, backend contract and test suite, and Playwright E2E**

Project: MediBytes ARD Hospital Demo

Repository: medibytesinternational-netizen/simple-demo

Source snapshot: simple-demo-main.zip supplied with the request

Report date: 06 Nov 2026

The latest user instruction was to keep the UI unchanged. Navigation, dashboard, and workflow UI edits made during the initial pass were reverted. The corresponding application source files were checked against the supplied archive and match byte-for-byte. This report records the existing interface, adds E2E tooling and documentation, and reports the validation that was actually run.

This is a software demo using synthetic records. It is not clinical-use guidance. No patient data or live model service was used for E2E checks.

1. Existing architecture and navigation

The archive contains the existing React and FastAPI application. No Git metadata was present, so source restoration was verified against the supplied ZIP instead of a Git diff.

Area	Existing implementation
Frontend	React 18, TypeScript, Vite, React Router, and TanStack Query.
Shell and navigation	Shell.tsx renders the legacy MenuBar. modules.ts supplies grouped links and route matching. Dock.tsx and Launchpad.tsx exist but are not mounted.
Routes	/dashboard, /patients, /patient, /timeline, /progress, /notes, /safety, /tasks, /documents, /discharge, /revenue, and /audit.
Backend	FastAPI and SQLAlchemy over SQLite. The contract check reports 16 modules (F1-F16) and 84 mounted API paths.
Human gates	Clinical draft approval, document confirmation, alert dismissal reason, task transitions, and discharge signing remain on existing human-controlled paths.

Before-and-after comparison

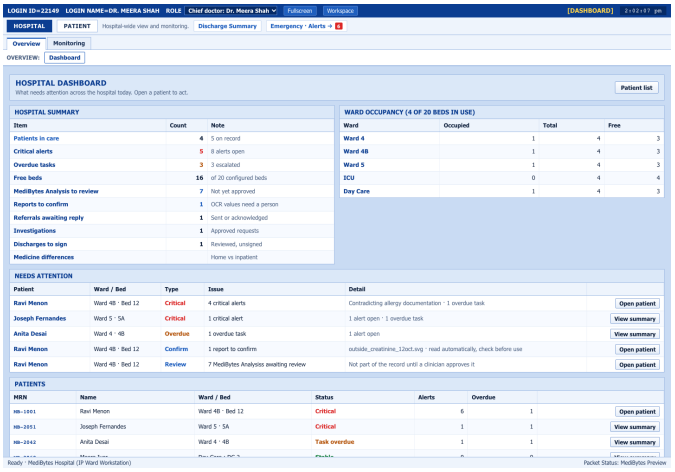
Measure	Before	After
Visible Dashboard navigation controls	7	7, unchanged
Duplicate quick-link destinations on patient routes	2	2, retained
Recommended Next Action card	Not present	Not added
F1-F16 backend modules	16	16, unchanged
Application UI source changes	Baseline	None; checked against supplied ZIP
Backend source changes	Baseline	None
Playwright E2E framework	Not installed	Added as a dev dependency

The unchanged 7-control count is measured on the Dashboard: two workspace tabs, two quick links, two section tabs, and one module link. On patient routes the quick Discharge Summary and Emergency Alerts links duplicate their section-menu destinations. They remain because the user asked for no UI changes.

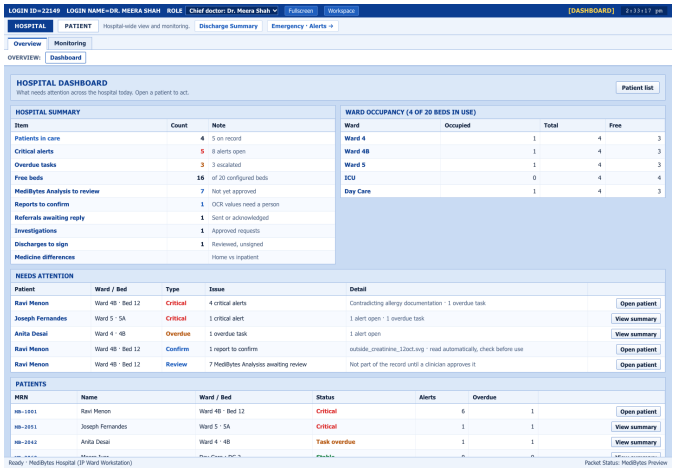
The existing menu exposes these routes through grouped sections. This work did not consolidate or remove those navigation controls.

2. Before-and-after screenshots

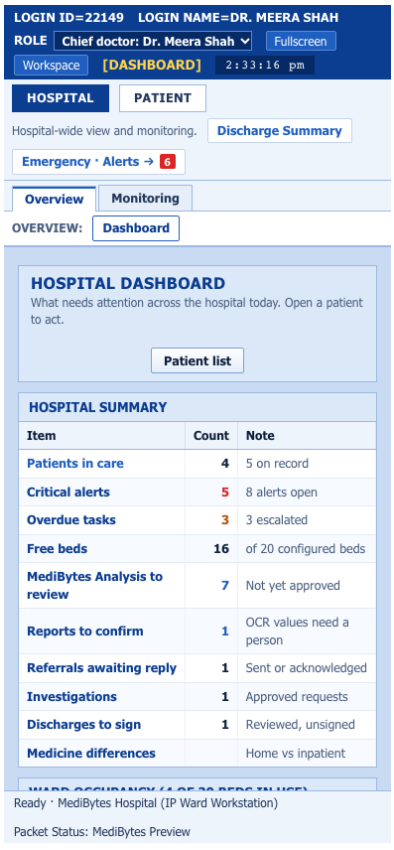
Both desktop images show the same existing Dashboard at 1440 x 1000. The after image was captured after restoring the UI source from the supplied ZIP. The matching appearance is intentional and was checked visually.



BEFORE - original Dashboard



AFTER - UI preserved



Mobile view - 390 x 844

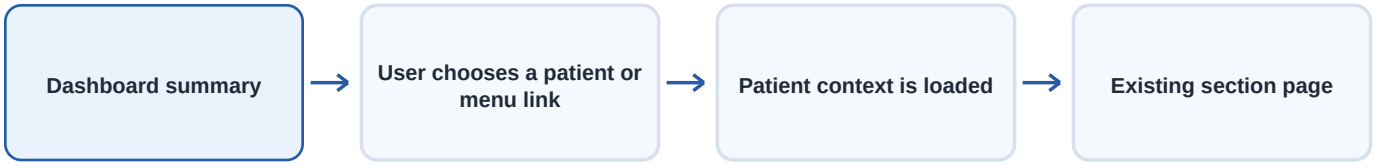
The original navigation wraps across two rows at this viewport. The E2E width check found no horizontal page overflow at 390 px. No responsive layout changes were made.

Observed fixture discrepancy: the Dashboard fixture reports 8 open alerts while the Safety page fixture reports 0 open alerts. The screenshots are synthetic preview data and are not evidence that those values agree. This inconsistency was documented, not changed.

3. Existing workflow maps

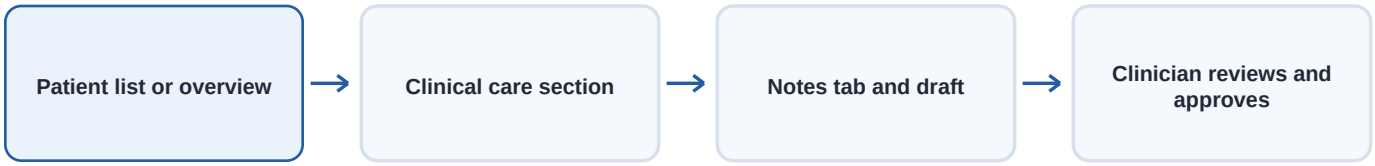
These diagrams document existing routes and human decisions. No automation was added. Arrows show navigation or a user-controlled transition.

A. Dashboard to an existing next destination



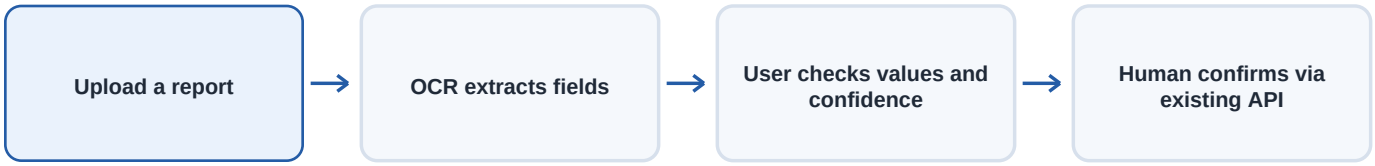
The current Dashboard shows counters and a needs-attention table. It has no ranked next-action CTA; the user selects an existing patient or menu destination.

B. Patient to clinical documentation



The selected patient is retained in session context. Draft review and approval remain in the existing review component and use the existing API.

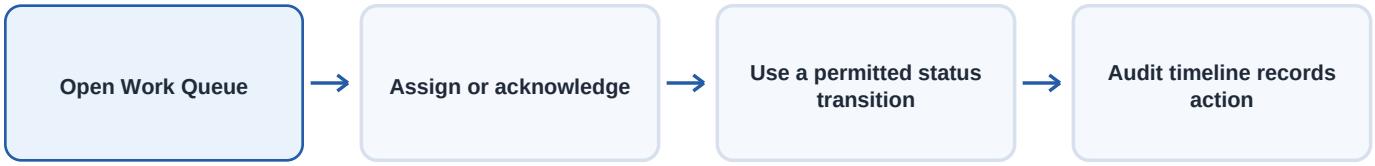
C. Document upload to human confirmation



Preview mode uses seeded synthetic documents and blocks writes. Confirmation was not performed in the E2E suite.

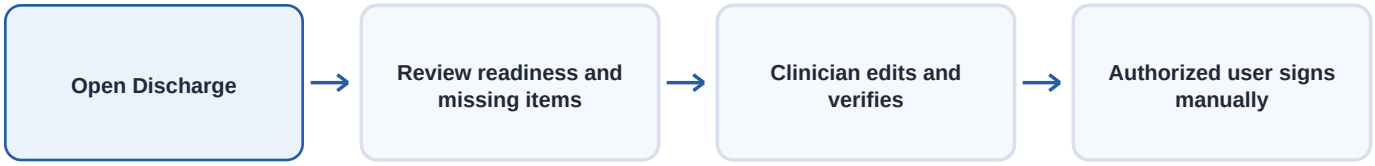
4. Existing queue and discharge workflows

D. Pending task to completion



Tasks retain assignment and audit fields through existing endpoints. E2E route checks did not perform task writes, assignment, escalation, or completion.

E. Discharge readiness to authorized signing



The existing discharge readiness and signing screens remain intact. No automatic approval or signature was added or exercised.

Existing safety gates

Workflow	Existing control retained
Clinical notes and drafts	Human review and approval use existing role and evidence checks.
Documents	OCR values require a human confirmation step.
Safety alerts	Dismissal requires a reason; alerts are not auto-dismissed.
Discharge	Final signing stays with an authorized human.
Billing	Claim review remains local; no external submission or payment occurred.

Functionality categories

Existing: routes, APIs, workflows, seeded records, review screens, and audit paths. **New:** Playwright E2E setup, validation documentation, screenshots, and this report. **Simulated:** cached preview data; preview writes are refused. **Not verified:** live backend responses, writes, role-specific approvals, external integrations, and actual discharge signing.

5. Test evidence and performance

Validation used the cached synthetic preview for browser checks and the existing backend contract and pytest suite. No API key or live model call was used.

Check	Result	Evidence
Frontend production build	PASS	npm run build; TypeScript and Vite build succeeded.
Python compile	PASS	python -m compileall -q app in an isolated Python 3.12 environment.
F1-F16 contract check	PASS	16 modules and 84 mounted paths.
Backend tests	PASS	659 passed in 27.22 seconds.
Playwright E2E	PASS	3 passed: 12 route targets, unchanged menu, patient-route duplicate links, responsive layout.
Browser requests and errors	PASS	0 failed requests, 0 page errors, and 0 /api HTTP requests in preview E2E.
Responsive widths	PASS	1440 x 1000, 768 x 1024, and 390 x 844; no page-level horizontal overflow.

Before-and-after runtime comparison

Measure	Baseline	Final	How measured
Cold Dashboard logical reads	17	17	Existing React Query definitions: shared metadata, patient state, dashboard, queues, and 9 feature draft reads.
Preview HTTP requests to /api/*	0	0	Playwright request listener; preview serves local fixture data.
New application runtime calls	0	0	No application source or query changes remain.
Frontend bundle JS / CSS	453.71 KB / 126.34 KB	453.71 KB / 126.34 KB	npm run build output; Playwright is dev-only.
New dependencies	0	1 dev dependency	@playwright/test; no production dependency added.

The 17 logical read count is derived from the existing shell and Dashboard query definitions. The 0 /api count is measured in preview mode and must not be interpreted as a live backend request count. Playwright's Chromium installation is a local test tool and is not included in the production bundle.

A first test iteration had a selector/count assertion mismatch. It was corrected; the final E2E run has 3 passed and 0 failed tests.

6. Files, limitations, and local commands

Files changed

README.md - added macOS/Linux backend, build, and E2E validation commands.

frontend/package.json and frontend/package-lock.json - added @playwright/test as a dev dependency and a test:e2e script.

frontend/playwright.config.ts - starts Vite in cached preview mode for E2E.

frontend/e2e/navigation.spec.ts - 3 tests for existing routes, unchanged menu, duplicate patient links, and responsive width.

reports/baseline-inventory.md and reports/screenshots/ - baseline notes and actual before/after screenshots.

reports/MediBytes_ARD_UI_Workflow_Improvement_Report.pdf - this report.

Limitations and unresolved items

- The UI simplification and new CTA automation were not implemented because the user directed that the UI must not change.
- The 2 duplicate quick links remain; navigation count remains 7 on the Dashboard.
- Preview fixture mismatch remains: dashboard summary says 8 open alerts; the Safety route says 0.
- The preview dashboard fixture omits medicine_discrepancies, so the existing table renders that field blank.
- No live backend, write path, authorization change, document upload, approval, discharge signature, or external integration was exercised.
- npm install reported 3 moderate dependency advisories. No audit fix or unrelated dependency upgrade was run.

Local commands

```
# Backend (from repository root)
cd backend
python3.12 -m venv .venv
.venv/bin/python -m pip install -r requirements-dev.txt
.venv/bin/python -m uvicorn app.main:app --reload --port 8000

# Frontend, in another terminal
cd frontend
npm install
npm run dev
# http://localhost:5173

# E2E (frontend directory; uses cached synthetic preview)
npx playwright install chromium
npm run test:e2e

# Validation
npm run build
../backend/.venv/bin/python -m pytest -q
```

Conclusion

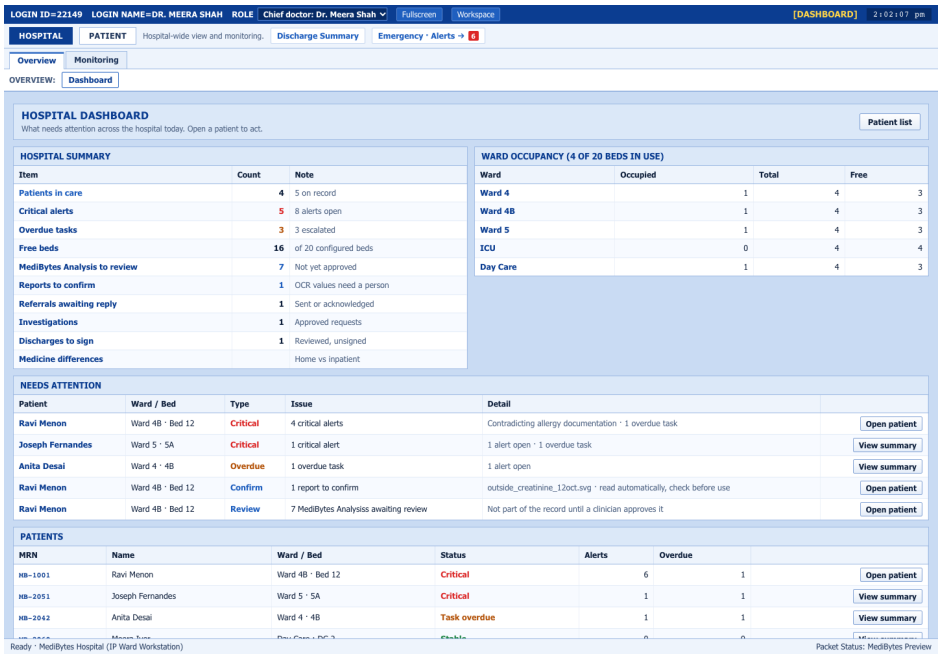
The existing UI and application behavior are unchanged. The project now has a reproducible cached-preview E2E suite, documented validation commands, baseline evidence, and a report. The requested navigation redesign and CTA automation remain unimplemented under the user's no-UI-change instruction.

Dashboard CTA and button reduction

Side-by-side screenshots from the supplied baseline and current build | same 1440 x 1000 viewport | synthetic preview data

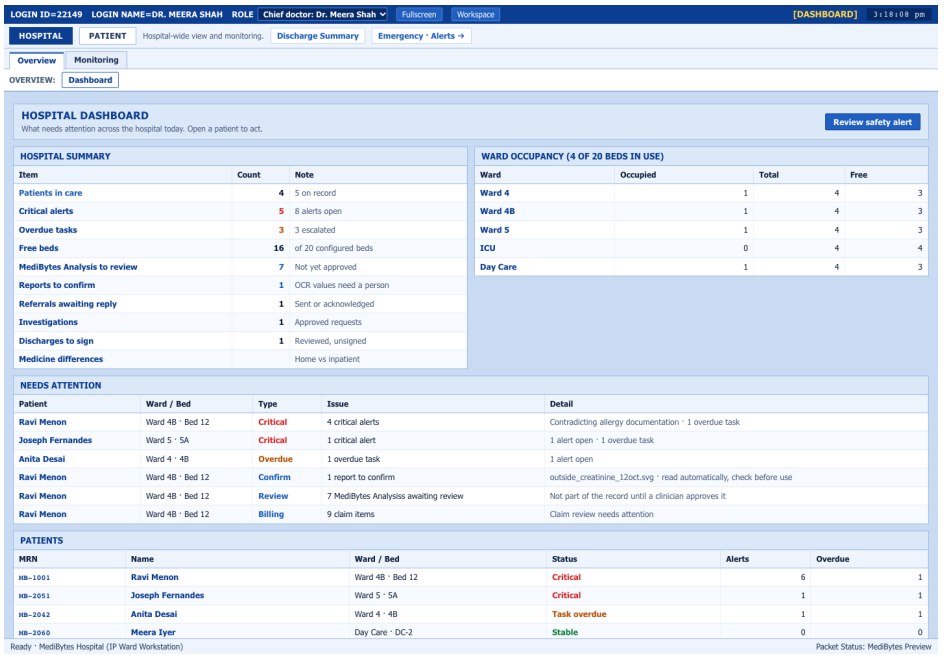
The current view keeps the existing blue workstation theme. Repeated row buttons are removed, the duplicate Patient list header action is replaced by one priority CTA, and patient names remain direct links to the patient chart or summary.

BEFORE - supplied baseline



Repeated Open patient / View summary buttons appear in both Dashboard tables. The header has a Patient list button.

AFTER - current build



One data-driven primary CTA remains. Patient names open records; repeated row buttons and the redundant header action are gone.

Change	Before	After
Patient page navigation	Discharge and Emergency Alerts appear as quick links and again in section menus.	Duplicate quick links are hidden on open patient pages; Clinical care and Safety & results menus retain access.
Contextual action	No Dashboard priority CTA.	CTA selects the highest-priority outstanding workflow and opens its existing route with patient context.
Theme	Blue workstation colors, type and component style.	Same colors, type and component style; no theme tokens changed.

Validation after the changes: frontend build passed; E2E passed 3 of 3 checks, covering the CTA route, patient-name navigation, duplicate-link removal, and responsive widths. Public static preview uses synthetic data; no /api requests were made.